WORKED SOLUTION

STEP I 2003, Question 4

Pure • Trigonometric Inequalities

> **STEP I 2003, Question 4**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/03-s1-q4

The key idea

The tempting move is to multiply both sides by $\cos\theta$, but $\cos\theta$ changes sign on $[0, 2\pi)$, and multiplying by a negative number reverses an inequality. So we split into the cases $\cos\theta > 0$ and $\cos\theta < 0$. One case turns out to be trivial, because $1 + \sin\theta$ is never negative. The other reduces to a single sine inequality once we write $\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$.

Solution

Write $f(\theta) = \dfrac{1 + \sin\theta}{\cos\theta}$. Throughout, $0 \le \theta < 2\pi$ and $\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$.

Case 1: $\cos\theta < 0$**, that is,** $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$

Since $\sin\theta \ge -1$, the numerator $1 + \sin\theta$ is at least 0, and the denominator is negative. So $f(\theta) \le 0 < 1$.

Every θ in $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ satisfies the inequality.

Case 2: $\cos\theta > 0$**, that is,** $0 \le \theta < \frac{\pi}{2}$ **or** $\frac{3\pi}{2} < \theta < 2\pi$

Multiplying by the positive number $\cos\theta$ keeps the direction of the inequality, so

$$f(\theta) \le 1 \iff 1 + \sin\theta \le \cos\theta \iff \sin\theta - \cos\theta \le -1.$$

Now $\sin\theta - \cos\theta = \sqrt{2}\left(\sin\theta\cos\frac{\pi}{4} - \cos\theta\sin\frac{\pi}{4}\right) = \sqrt{2}\sin\left(\theta - \frac{\pi}{4}\right)$, so the condition becomes

$$\sin\left(\theta - \tfrac{\pi}{4}\right) \le -\tfrac{1}{\sqrt{2}}.$$

Ask yourself Which values of $\phi = \theta - \frac{\pi}{4}$ do we need to look at?

We need only the two ranges allowed in this case:

- If $0 \le \theta < \frac{\pi}{2}$, then $-\frac{\pi}{4} \le \phi < \frac{\pi}{4}$. On this interval $\sin\phi$ is increasing and $\sin\phi \ge \sin(-\frac{\pi}{4}) = -\frac{1}{\sqrt{2}}$, with equality only at $\phi = -\frac{\pi}{4}$. So the only solution here is $\theta = 0$. (Check: $f(0) = 1/1 = 1$, so equality holds.)
- If $\frac{3\pi}{2} < \theta < 2\pi$, then $\frac{5\pi}{4} < \phi < \frac{7\pi}{4}$. On this open interval $\sin\phi < -\frac{1}{\sqrt{2}}$, since $\sin\phi$ equals $-\frac{1}{\sqrt{2}}$ at both ends and dips to -1 at $\phi = \frac{3\pi}{2}$ in between. So *every* θ in $\left(\frac{3\pi}{2}, 2\pi\right)$ is a solution.

Putting it together

The value $\theta = \frac{3\pi}{2}$ is excluded because $\cos\theta = 0$ there. Combining the cases, the inequality holds exactly when

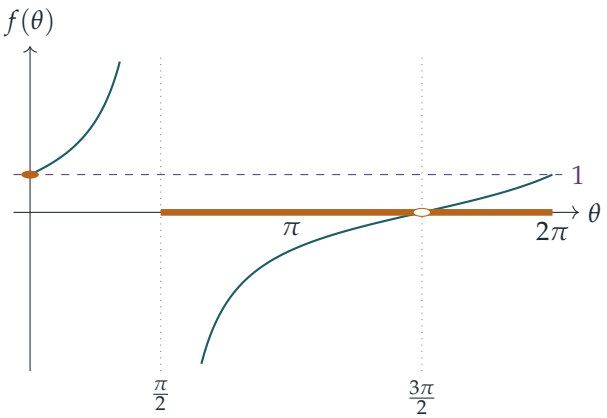
$$\theta = 0 \qquad \text{or} \qquad \frac{\pi}{2} < \theta < \frac{3\pi}{2} \qquad \text{or} \qquad \frac{3\pi}{2} < \theta < 2\pi.$$

A slicker view

Using $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$, $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$ and $1 = \cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}$,

$$f(\theta) = \frac{\left(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\right)^2}{\left(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\right)\left(\cos\frac{\theta}{2} - \sin\frac{\theta}{2}\right)} = \frac{1 + \tan\frac{\theta}{2}}{1 - \tan\frac{\theta}{2}} = \tan\left(\frac{\theta}{2} + \frac{\pi}{4}\right),$$

cancelling the common factor, which is nonzero because $\cos\theta \neq 0$. (At $\theta = \pi$ the middle expression is undefined, but the first and last are equal there too.) As θ runs over $[0, 2\pi)$, $\psi = \frac{\theta}{2} + \frac{\pi}{4}$ runs over $[\frac{\pi}{4}, \frac{5\pi}{4})$. The graph of $\tan\psi$ shows at once that $\tan\psi \le 1$ only at $\psi = \frac{\pi}{4}$ and for $\frac{\pi}{2} < \psi < \frac{5\pi}{4}$. This is the same answer, once we remove $\psi = \pi$ (that is, $\theta = \frac{3\pi}{2}$).



The solution set (amber) is where the curve lies on or below the dashed line $f = 1$. Note the hole at $\theta = \frac{3\pi}{2}$: there $f(\theta) \to 0$, but f is not defined.

Checks

- $\theta = 0$: $f(0) = 1$, on the boundary. Just above 0 the numerator exceeds 1 while the denominator is below 1, so $f > 1$. This confirms that $\theta = 0$ is an isolated solution.
- $\theta = \frac{7\pi}{4}$: $f = \left(1 - \frac{1}{\sqrt{2}}\right) / \frac{1}{\sqrt{2}} = \sqrt{2} - 1 \approx 0.414 \leq 1$. Good.
- $\theta = \pi$: $f = 1/(-1) = -1 \leq 1$. Good.
- We tested 2×10^5 random θ in $[0, 2\pi)$ and found complete agreement with the stated solution set. Near $\theta = \frac{3\pi}{2}$ we found $f \approx 5 \times 10^{-5}$, consistent with $f \to 0$.

Common slips

- Multiplying through by $\cos\theta$ without splitting into cases. This gives $1 + \sin\theta \leq \cos\theta$ everywhere, which wrongly throws away the whole interval $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$.
- Forgetting the isolated solution $\theta = 0$. The question allows $\theta = 0$, and equality counts.
- Including $\theta = \frac{3\pi}{2}$. The expression is $\frac{0}{0}$ there, so it is not defined, even though its limit is 0.

Beyond STEP

The lesson is that inequalities are only preserved by *increasing* operations: multiplying by a positive number, applying an increasing function, and so on. Keeping track of this is the daily business of real analysis. The removable singularity at $\theta = \frac{3\pi}{2}$, where f has a limit but no value, is another theme there: continuity is about limits agreeing with values. Abbott's *Understanding Analysis* develops both ideas carefully and readably.

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